

Yen-Teng Chang

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EDUCATION

2024–2026 **M.S., Civil Engineering, Structural Engineering Division**

National Taiwan University, Taipei, Taiwan

Thesis: *A Spatial Computing and Edge AI Framework for Automated Indoor Defect Inspection*

Advisor: Prof. Chia-Ming Chang

Thesis grade: **A+** | DOI: 10.6342/NTU202603079

2020–2024 **B.S., Civil Engineering**

National Taiwan University, Taipei, Taiwan

PROFESSIONAL DEVELOPMENT

Research Assistant, *Department of Civil Engineering, National Taiwan University* (June 2026 – Present)

- ◇ Develop AI-based inspection systems consisting of model training, on-device deployment, and client delivery across rail, aviation, and building infrastructure
- ◇ Author technical proposals and system specifications for industry-funded projects

Architecture-Agnostic Knowledge Distillation for On-Device Defect Detection, *Master's thesis research, National Taiwan University* (2024–2026)

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- ◇ Developed an architecture-agnostic dual-path knowledge-distillation framework that transfers complementary supervision from RT-DETR-L and YOLO11-L teachers to a 2.6M-parameter YOLO11n student through cross-architecture pseudo-label supervision and detection-head alignment.
- ◇ Improved the student model's mAP50–95 by 0.026 while preserving its inference architecture, enabling edge deployment with no additional inference-time parameters or computation.

Auto Inspector — Automated Building Defect Inspection and Digital Twin System, *Sinotech Engineering Consultants R&D contract*, Sole developer (2025–2026)

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- ◇ Developed a fully on-device iPad inspection and digital-twin system integrating LiDAR–IMU–RGB SLAM, RoomPlan, AR raycasting, and custom mesh processing to localize detected defects in a shared 3D coordinate frame without external hardware, remote computing, or network connectivity.
- ◇ Delivered a self-contained field workflow generating 3D models, floor plans, inspection records, and coordinate-aware defect data, with <1.5% geometric error and an 85% reduction in inspection time.

Lock-Up Device (LUD) Intelligent Inspection — Deterioration Segmentation and Standards-Based Condition Assessment, *Sinotech Engineering Consultants R&D project involving Taiwan Railway* (Sole developer, November 2024–Present)

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- ◇ Developed and field-deployed an end-to-end LUD inspection system integrating deterioration segmentation, geometric quantification, and standards-based condition assessment for an active railway engineering project.
- ◇ Designed a standards-constrained vision-language model (VLM) that restricts assessment outputs to bridge-inspection requirements through structured decoding, rule validation, confidence gating, and auditable decision logs, coupled with a graded human-in-the-loop workflow for automatic acceptance, manual confirmation, and mandatory review.

Crack Monitoring System — Real-Time Multi-Camera Deployment, Sanlien Technology Corp. (Sole developer, September 2025 – January 2026)

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- ◇ Developed a multi-camera crack-monitoring system that automatically quantifies crack length and maximum width from segmentation masks across concurrent RTSP streams, with physical-scale calibration and continuous data logging.
- ◇ Designed a modular desktop application and packaged it as a self-contained Windows executable for direct client use, providing system-status, monitoring-wall, and detailed camera views; integrated continuous data logging and a rolling buffer with the host seismic-detection system to automatically archive synchronized crack measurements and imagery from before, during, and after each seismic event.

Site Engineering Intern, Taoyuan International Airport Terminal 3 Project — Samsung C&T × RSEA Engineering; construction supervision by CECI Engineering Consultants (June 2023 – September 2024)

- ◇ Produced shop and site drawings; managed site operations, inspection, and construction scheduling
- ◇ Resolved a construction conflict in a large X-shaped pile frame with the chief structural engineer

Site Engineering Intern, Chuan Gao Engineering Co., Ltd. (June 2023 – September 2024)

- ◇ Prepared site construction drawings, reconciled design documents with field practice, and supervised construction and scheduling across a triple-rail station joint development, a residential complex, and Ministry of National Defense barracks.
- ◇ Automated most of the drafting workflow with AutoLISP and Excel VBA; the tools remained in production use after the internship.

Structural Engineering Intern, Elements Structure, Taipei, Taiwan. (June 2023 – September 2023)

- ◇ Performed structural modeling, drafting, and analysis across three building projects: large-scale structural design, skybridge detailing/modeling, and composite pile stiffness computation

PROFESSIONAL LICENSURE

Licensed Civil Engineer, Taiwan, 2025

- ◇ **Ranked 65th of 3,810 candidates nationwide (top 1.7%; acceptance rate: 16.26%)**

AWARDS

2nd Place, Cathay Housethon BIM Competition, 2022

- ◇ Led building model design, drafting, model construction, and 3D rendering for the full entry

PUBLICATIONS AND PRESENTATIONS

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Journal

- ◇ **Chang, Y.-T.**, and Chang, C.-M. (2026). “A Spatial Computing and Edge AI Framework for Automated Indoor Defect Inspection.” To be submitted to *Engineering Applications of Artificial Intelligence*.
- ◇ Togi, R. S., **Chang, Y.-T.**, and Chang, C.-M.* (2026). “A Mobile AR Application Integrating SLAM and Deep Learning for Indoor Structural Inspection.” *Smart Structures and Systems*, under review.

Conference

- ◇ **European Workshop on Structural Health Monitoring (EWSHM 2026)**, SHM in Action session (presentation with live system demonstration), Toulouse, France, 7–10 July 2026.
- ◇ **Chang, Y.-T.**, and Chang, C.-M. (2026). “A Spatial Computing and Edge AI Framework for Automated Building Defect Inspection.” *The 18th National Conference on Structural Engineering / The 8th National Conference on Earthquake Engineering*, Taiwan, 19–21 August 2026.

TECHNICAL SKILLS

- ◇ **Programming:** Python, Swift, C++, Java, MATLAB
- ◇ **Deep Learning and AI:** YOLO (classification, detection, segmentation), RT-DETR, knowledge distillation, instance and semantic segmentation, model optimization and deployment, Core ML, RAG, LLM and VLM integration
- ◇ **Spatial Computing and Mobile:** ARKit, RoomPlan, SpriteKit, iOS application development, LiDAR point cloud processing, Raycast coordinate anchoring
- ◇ **Structural Analysis and Design:** ETABS, SAP2000, SAFE, Midas, TEASPA
- ◇ **Drafting and BIM:** AutoCAD, RCAD, Revit, Navisworks, SketchUp
- ◇ **Systems:** Linux shell, SSH, Docker, CLI, GPU cluster training workflow, CUDA, cross-platform macOS and Windows development

RESEARCH INTERESTS

- ◆ **Structural health monitoring, multimodal sensing, and uncertainty-aware condition assessment**
- ◆ **Edge AI and efficient knowledge transfer for field-deployed infrastructure inspection**
- ◆ **Spatial computing, defect modeling, and digital twins for infrastructure assessment**
- ◆ **Standards-constrained AI and human-in-the-loop engineering decision support**